

Henry Luo

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LinkedIn: [linkedin.com/in/henry-luo-065711233/](https://www.linkedin.com/in/henry-luo-065711233/) | **Portfolio:** the-other-hello-there.github.io/HenryLuo-Portfolio/
Summary: Industry 5.0 actionist with mechanic experience specialized in turning fragile manual processes into repeatable systems. Experienced in combining first-principles-driven design, systematic TRIZ methodologies, and practical manufacturing into mindful engineering solutions.

EDUCATION

BSE Mechanical Engineering, BS Aeronautical Engineering Technology

Aug 2021 – December 2026

Purdue University | West Lafayette, IN | GPA: 3.72/4

- **Involvement:** Purdue Aerial Robotics Team (R&D Mechanical Systems), Humanoid Robotics Club (Calf & Leg 2), American Society of Mechanical Engineers (Small Projects Competition), Purdue Aviation Day (Ramp Volunteer), All-American Marching Band (1st Trombonist)

CERTIFICATIONS

Fundamentals of Engineering Exam Pass – NCEES

September 2026

Airframe & Powerplant Mechanic Certificate – FAA

April 2025

Certified SOLIDWORKS Associate – VirtualTester Online

May 2022

Creo User Certificate – PTC University

April 2026

Non-Destructive Testing Certificate (PTII, MTII, ETI) – Purdue University SATT

December 2021

Group 2 Human Research Certification – CITI Program

January 2026

Python Certificate – Purdue University EBEC

May 2022

EXPERIENCE

R&D Manufacturing Engineering Co-Op

Jun 2025 – Present

Midwest Aerospace Casting, LLC | Merrillville, IN

- **SCADA Architecture & Standardization:** Architected the SCADA backend for data acquisition & automated analysis to support production decision-making & future Industry 5.0 growth. Created training modules, reusable templates, & dataset tools to expand internal capability for ongoing development.
- **IIOT Development:** Led end-to-end development of 3 IIOT devices, ensuring survivability in harsh foundry environments, SCADA connectivity, & minimal operator load. Mentored intern IIOT projects.
- **Tooling Support:** Anchored FDM & MJP 3D printing operations by supporting DFAM for custom tooling/components & printer upkeep. Developed custom core dryers, core boxes, & machine upgrades.
- **New Product Development:** Designed gating systems for single-source aerospace components & prepared manufacturing operational documentation for new products.
- **Process Improvement:** Analyzed current & future state metrics during a 3P event, utilizing Takt times & SWIP counts to design new facility requirements that meet customer demand & maintain cross-training. Proved a 40% efficiency increase potential for the assembly process during a Kaizen event.
- **Defect Resolution:** Performed first-principles RCCA investigations to resolve recurring casting defects.

Manufacturing Engineering Intern

Jun 2022 – Aug 2022

Applied Composite Structures, Inc. | Brea, CA

- **Process Organization:** Redesigned workstation layouts & material traveling patterns to reduce part loss.
- **Production Benchmarking:** Collaborated with a technician on Gemba Walks to benchmark present production performance with value stream maps, time trials, takt times, & spaghetti diagrams.

Programming Intern

Jun 2020 – Aug 2020

SmartAxiom, Inc. | Brea, CA

- **Network Simulation:** Simulated an IoT network using named & UDP pipes in Bash & Python.

SKILLS

Design & Analysis: CAD (SolidWorks, NX, Inventor, CATIA, Creo, Fusion, Onshape), Mechanical Engineering, Iterative Design, TRIZ, FEA, MATLAB, Microsoft Office Suite

Manufacturing: 3D Printing (FDM, MJP), Sand Casting Manufacturing (Magnesium, Aluminum), Composite Manufacturing (Prepreg), CAM, GD&T, Lean Six Sigma, FMEA/RCCA, LCCA, Technical Documentation

IoT & Automation: Programming (Python, C, Java, Arduino), SCADA (Ignition), Cybersecurity

Projects

Ghost Arm | StarkHacks (Best Implementation of Viam)

April 2026

- **Rapid Development:** Developed a tele-operational robotic arm in 36 hours to improve manufacturing and maintenance ease and safety in “dark, dirty, and dangerous” environments.

Felon the Freestylin’ Feline | ME 444 Interactive Toy (Honorable Mention)

May 2026

- **Assembly Management:** Developed a split chassis to isolate aesthetic pieces from functional components while ensuring ease of 3D-print fabrication, revision, & assembly.

Hat Tipper | ME 444 Battle Bots (1st Place)

March 2026

- **Practical Innovation:** Developed a defense mechanism resembling a hat to lift and immobilize opponents.

Small Projects | ASME RC Kart Design & Race (1st Place)

May 2024

- **Cross-Functional Prototyping:** Collaboratively designed and manufactured a mini (6”x10”) 3D-printed RC kart with 4-wheel suspension, dogbone transmission, and linkage steering, completing 3 laps in 2:21.